

LAPORAN PRAKTIKUM
Deploy Server Layanan Internal



Transformasi Digital

Dosen Pengampu:

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BAB I - Pendahuluan

Latar Belakang:

Deployment server merupakan proses menyiapkan layanan jaringan agar dapat digunakan secara optimal. Pada praktikum ini dilakukan implementasi File Server, Database Server, Web Server, dan Backup Server menggunakan Ubuntu Server.

Tujuan:

1. Menginstal Ubuntu Server.
2. Menginstal dan mengonfigurasi Nginx.
3. Menginstal dan mengonfigurasi MySQL.
4. Menginstal dan mengonfigurasi Samba.
5. Melakukan backup data menggunakan Rsync.
6. Melakukan pengujian seluruh layanan.

BAB II - Dasar Teori

Ubuntu Server adalah sistem operasi Linux untuk kebutuhan server.

Nginx berfungsi sebagai Web Server.

MySQL berfungsi sebagai Database Server.

Samba digunakan sebagai File Server.

Rsync digunakan untuk proses backup data.

BAB III - Alat dan Bahan

Hardware:

- Laptop/PC
- RAM minimal 4 GB

Software:

- Oracle VirtualBox
- Ubuntu Server 24.04 LTS
- Nginx
- MySQL Server
- Samba
- Rsync
- SSH

BAB IV - Langkah Praktikum

1. Login ke Ubuntu Server.

```
Ubuntu 24.04.4 LTS Panji tty1
```

```
Panji login: panji  
Password: _
```

2. Melihat informasi sistem menggunakan hostnamectl.

```
panji@Panji:~$ hostnamectl  
      Static hostname: Panji  
            Icon name: computer-vm  
          Chassis type: vm  
        Machine ID: 190e5e03f1ba47169b5c088c97cb1ebf  
         Boot ID: 8b5efee89cb04902ae476e04a63db3fc  
    Virtualization: oracle  
  Operating System: Ubuntu 24.04.4 LTS  
           Kernel: Linux 6.8.0-117-generic  
    Architecture: x86_64  
   Hardware Vendor: innotek GmbH  
   Hardware Model: VirtualBox  
  Firmware Version: VirtualBox  
    Firmware Date: Fri 2006-12-01  
   Firmware Age: 19y 6month 4w  
panji@Panji:~$
```

3. Menampilkan alamat IP menggunakan hostname -I.

```
panji@Panji:~$  
Display all 163 possibilities? (y or n)  
panji@Panji:~$  
Display all 163 possibilities? (y or n)  
panji@Panji:~$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UN  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq  
    link/ether 08:00:27:cb:9b:28 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.18.241/24 metric 100 brd 192.168.18.255 scope  
        valid_lft 3458sec preferred_lft 3458sec  
    inet6 2404:8000:1013:1e88:a00:27ff:feeb:9b28/64 scope glob  
        valid_lft 259058sec preferred_lft 172658sec  
    inet6 fe80::a00:27ff:feeb:9b28/64 scope link  
        valid_lft forever preferred_lft forever
```

4. Update repository (sudo apt update).

```
panji@Panji:~$ sudo apt update  
[sudo] password for panji:  
Hit:1 http://archive.ubuntu.com/ubuntu noble InRelease  
Hit:2 http://security.ubuntu.com/ubuntu noble-security InRelease  
Hit:3 http://repository.netdata.cloud/repos/edge/ubuntu noble/ InRelease  
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease  
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease  
Hit:6 http://repository.netdata.cloud/repos/repoconfig/ubuntu noble/ InRelease  
Reading package lists... 5%
```

5. Install Nginx dan cek status service.

```
panji@Panji:~$ sudo apt install nginx -y
[sudo] password for panji:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  nginx-common
Suggested packages:
  fcgiwrap nginx-doc
The following packages will be upgraded:
```

6. Install MySQL dan lakukan pengujian SHOW DATABASES.

```
panji@Panji:~$ sudo systemctl restart sudo
panji@Panji:~$ sudo apt install mysql-server -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following packages were automatically installed and are no longer required:
  galera-4 libconfig-inifiles-perl libdbd-mysql-perl libdbi-perl libmariadb3 libmysqlclient21 libwrap0
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
  libevent-pthreads-2.1-7t64 libmecab2 libmysqlclient21 libprotobuf-lite32t64 mecab-ipadic mecab-ipadic-utf8
```

7. Install Samba dan konfigurasi smb.conf.

```
panji@Panji:~$ sudo apt install samba -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
samba is already the newest version (2:4.19.5+dfsg-4ubuntu9.6).
```

8. Konfigurasi firewall menggunakan UFW.

```
panji@Panji:~$ sudo ufw status
Status: inactive
panji@Panji:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs            291M  5.2M  285M   2% /run
/dev/sda2        25G   6.9G   16G  31% /
tmpfs            1.5G   0   1.5G   0% /dev/shm
tmpfs            5.0M   0   5.0M   0% /run/lock
Downloads       476G  440G   36G  93% /media/sf_Downloads
tmpfs            290M  12K  290M   1% /run/user/1000
```

9. Melakukan backup menggunakan Rsync.

```
panji@Panji:~$ sudo rsync -av /srv/shares/ /backup/
sending incremental file list
created directory /backup
./
admin/
dokumen/
dokumen/test.txt.txt
riset/
```

10. Menguji seluruh layanan service

Menguji Layanan NGINX, MYSQL, dan SAMBA menggunakan sudo systemctl

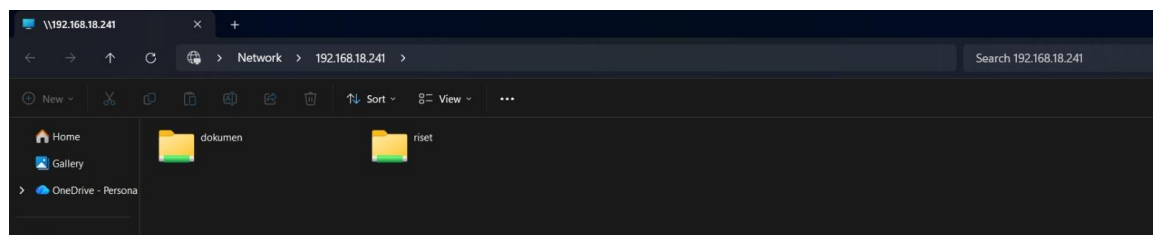
```
Process: 108268 ExecCondition=/usr/share/samba/is-configured smb (code=exited, status=0/SUCCESS)
Main PID: 108271 (smbd)
Status: "smbd: ready to serve connections..."
Tasks: 3 (limit: 3382)
Memory: 7.6M (peak: 7.6M)
CPU: 109ms
CGroup: /system.slice/smbd.service
├─108271 /usr/sbin/smbd --foreground --no-process-group
├─108277 "smbd: notifyd"
└─108278 "smbd: cleanupd"

un 30 07:24:16 Panji systemd[1]: Starting smbd.service - Samba SMB Daemon...
un 30 07:24:16 Panji (smbd)[108271]: smbd.service: Referenced but unset environment variable evaluates to an empty string: SMBOPTIONS
un 30 07:24:16 Panji systemd[1]: Started smbd.service - Samba SMB Daemon.
anjie@Panji:~$ systemctl status mysql
mysql.service - MySQL Community Server
Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
Active: active (running) since Tue 2026-06-30 07:24:19 UTC; 1min 39s ago
Process: 108288 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
Main PID: 108288 (mysqld)
Status: "Server is operational"
Tasks: 38 (limit: 3982)
Memory: 364.3M (peak: 370.5M)
CPU: 5.244s
CGroup: /system.slice/mysql.service
├─108288 /usr/sbin/mysqld

un 30 07:24:16 Panji systemd[1]: Starting mysql.service - MySQL Community Server...
un 30 07:24:19 Panji systemd[1]: Started mysql.service - MySQL Community Server.
anjie@Panji:~$ systemctl status nginx
nginx.service - A high performance web server and a reverse proxy server
Loaded: loaded (/usr/lib/systemd/system/nginx.service; disabled; preset: enabled)
Active: active (running) since Tue 2026-06-30 07:24:16 UTC; 1min 56s ago
Docs: man:nginx(8)
Process: 108224 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master process on; (code=exited, status=0/SUCCESS)
Process: 108231 ExecStart=/usr/sbin/nginx -g daemon on; master process on; (code=exited, status=0/SUCCESS)
Main PID: 108234 (nginx)
Tasks: 5 (limit: 3382)
Memory: 3.8M (peak: 4.3M)
CPU: 53ms
CGroup: /system.slice/nginx.service
├─108234 "nginx: master process /usr/sbin/nginx -g daemon on; master process on;"
├─108235 "nginx: worker process"
├─108236 "nginx: worker process"
├─108237 "nginx: worker process"
└─108238 "nginx: worker process"

un 30 07:24:16 Panji nginx[108224]: 2026/06/30 07:24:16 [warn] 108224#108224: conflicting server name "_" on 0.0.0.0:80, ignored
un 30 07:24:16 Panji nginx[108231]: 2026/06/30 07:24:16 [warn] 108231#108231: conflicting server name "_" on 0.0.0.0:80, ignored
anjie@Panji:~$
```

Folder IP server berhasil dibuka dari windows



BAB V - Hasil Praktikum

Seluruh layanan server berhasil diimplementasikan dan diuji. Web Server menggunakan Nginx, Database Server menggunakan MySQL, File Server menggunakan Samba, serta

Backup Server menggunakan Rsync dapat berjalan sesuai fungsi. Tambahkan screenshot pada setiap tahapan praktikum sebagai bukti implementasi.

BAB VI - Kesimpulan

Deployment server layanan internal berhasil dilaksanakan menggunakan Ubuntu Server. Konfigurasi layanan Nginx, MySQL, Samba, dan Rsync menunjukkan bahwa server mampu menyediakan layanan web, database, file sharing, serta backup data dengan baik.

Daftar Pustaka

1. Ubuntu Server Documentation
2. Nginx Documentation
3. MySQL Reference Manual
4. Samba Documentation
5. Rsync Documentation